

CS 471 Team 2 Final Project Proposal
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Motivation: We seek to create a model which can determine if a bank transaction is fraudulent based on data collected at the time. The following prior study had the same motivation: <https://www.kaggle.com/code/kishankumar2110/fraudulent-transactions-predict-with-99-accuracy>. However, we believe a smaller, simpler model can be built by only including a subset of all the features. This new model will achieve a similar accuracy to that of the study. Also, we found an article on Google Scholar that actually uses the same dataset from kaggle named “fraudulent transaction” and they mention using Decision tree, Logistic Regression, and Random Forest algorithms: <https://philpapers.org/rec/MEGFFT-2>. Before finding this article, we believed oversampling the data would be necessary as the dataset originally is very unbalanced and the article even states that “On behalf of a rather unbalanced dataset oversampling was required. After oversampling, 60% legal and 40% unlawful transactions are found.” (pg. 31).

Dataset: Our dataset is the fraudulent transactions dataset (<https://www.kaggle.com/datasets/vardhansiramdasu/fraudulent-transactions-prediction>). The dataset is composed of a series of simulated bank transactions between different individuals. The twelve features included in it, cover the transaction amount as well as the balances of the respective accounts. The dataset is formatted appropriately, with each column only containing one element. However, a significant imbalance is present between fraudulent transactions and non-fraudulent transactions. Pre-processing would involve utilizing over-sampling to ensure fraudulent transactions are equally represented.

Method: Currently, we foresee using a statistical analysis approach to differentiate fraudulent transactions using the deposit/ withdrawal amount and the balance of the accounts. Comparing a transaction’s respective features against mean or median values obtained from the totality of the data will achieve this. In addition to a rules-based method, we plan to create a decision tree machine learning model which can determine between fraudulent transactions.

Intended experiments: In both approaches, we will seek to eliminate low information features from being taken into considering. For the rules-based model, we will experiment with various thresholds of deviation from the mean. During the experimentation phase of the machine learning model, it is likely the maximum height of the tree and leaf amount will be adjusted.

Model evaluation: To evaluate our models, we will use the recall metric. Recall gives more credence to false negatives. From the perspective of a bank, one would care more about fraudulent transactions that were not flagged as such rather than non-fraudulent transactions

being mistakenly labeled as the former. As such, using the recall metric to evaluate our model will ensure this view is adhered to.